

TIM 172A: MOT-I, Lecture, 10/2/25

Agenda:

1. Organize the readings [Lecture Notes for Competitive Strategies (LNCS), available on Canvas; The Disk Drive Industry]
2. Introduction to Functional Maps
(HW#2, Prob 1: The Disk Drive Industry)
↑ also on Canvas
3. Main topic: Industry Analysis
- Personal Computer Industry
(HW#2, Prob. 2: INTEL)
↑ handout on Canvas
4. HW#2 has been posted: due Thursday, next week

TIM 172A is an engineering course:

- Please do not submit essays as solutions
- Structure your work to maximize information content: readability, user-friendly

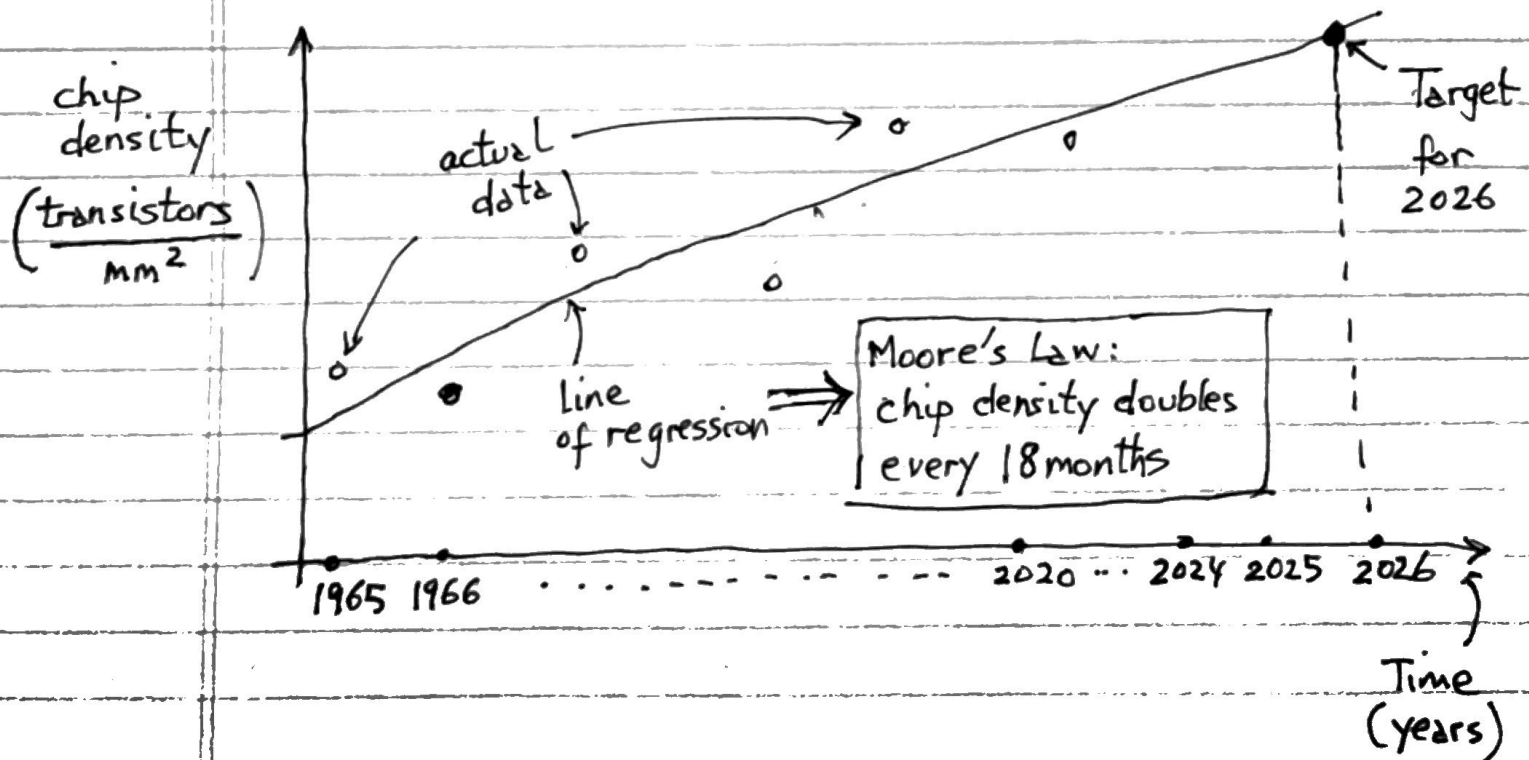
2. Functional Maps : An introduction

Functional maps can come in many forms:
graph, table, pictorial

Q : How does a tech. company (e.g., Intel, Seagate,..) set technical (product) performance targets for what they plan to do for the next year? for the next 3 years?,

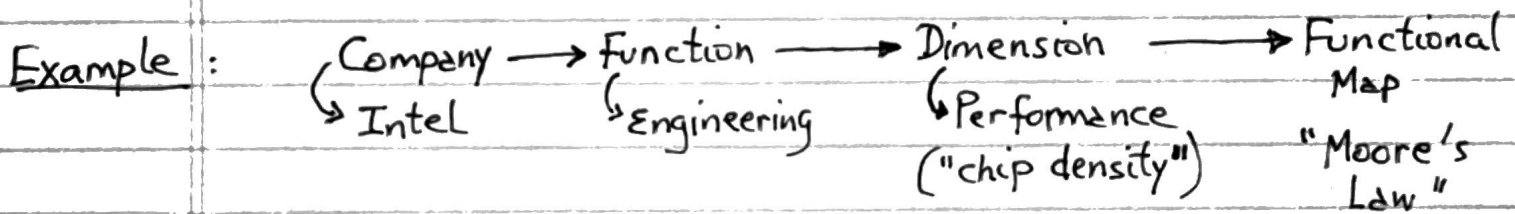
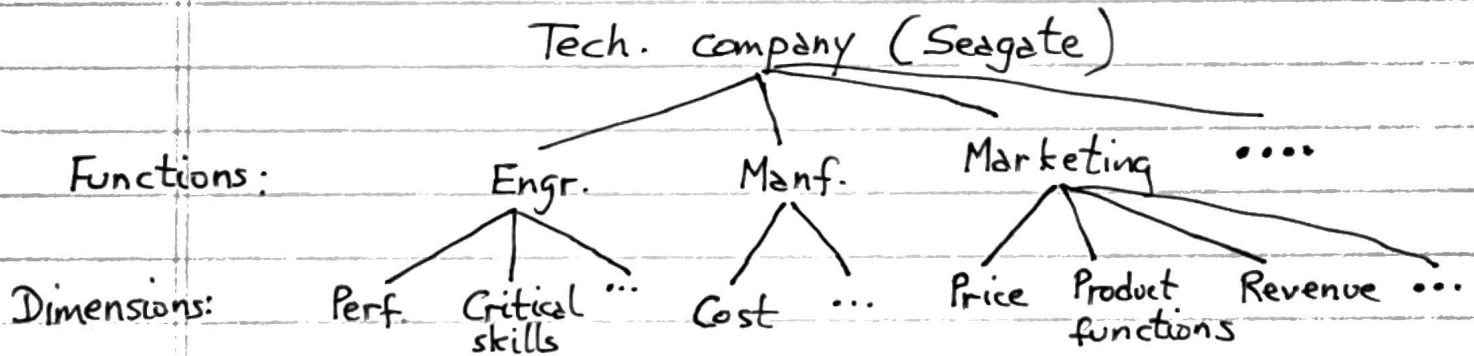
A : By creating the appropriate functional maps, that enable the prediction (or forecasting) of the appropriate tech. or product performance parameter.

The most famous functional map in the IT industry is Moore's Law.



Definitions:

- (1) Function refers to organizational functions:
Engineering, Manufacturing, Marketing,
Finance, ...
- (2) Each of these functions has important dimensions:
For example, the key dimensions of engineering are product performance, critical skills, ...

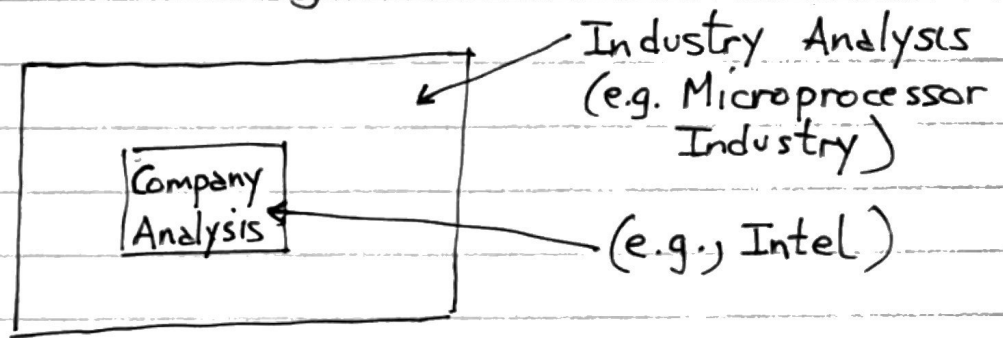


You are now ready to attempt

HW # 2, Prob. 1 : The Disk Drive Industry

2. Organizing the readings (on Canvas)
- 2.1. Read the "Lecture Notes for Competitive Strategy ..." (LNCS)

Guide for reading LNCS



Topics in LNCS	Section/Pages	Lecture #
1. Process (or Plan or Approach)	→ Pages 17-19 (Section 5)	
- Company Analysis		→ L# 4 (next Tue)
- Industry Analysis		→ L# 3 (Intro)
- Assessment/Evolution		→ L# 4 (Details)
2. Industry Analysis	Section 3 (Pages 11-15) 5-step process	Main topic for today
3. Functional Maps	Section 4 (Page 15)	L# 3 (Intro) L# 4 (Details)

- 2.2 Read the "Intel Case Study" (on Canvas)
- Use the SPSP to solve HW# 2, Prob # 2
- Focus on the Industry Analysis for now

3. Structural Analysis of Industry (aka Competitive Strategy Analysis)

Competitive Strategy comes from "The Positioning School" (Porter $\approx$ 1980)

Terminology :

Industry : The set of players for some technology or product type, e.g., disk-drive industry, consumer electronics, health-care industry,

Players (in the industry) : Competitors, new entrants, substitutes, suppliers

Market : Buyers (or customers) of the product/service

Forces : The micro-economic forces that exist among the players.

Company (or Firm or Enterprise) : is embedded in the industry of interest

Step 1 of the SPSP : Definition of the problem

Porter asked the following questions

- How attractive is an industry from the point of view of the players?
- What position or competitive strategy should a competitor (in the industry) adopt?

Example : The Personal Computer Industry $\approx$ 1980s

Step 2 of the SPSP: Plan or Process :

5-step process
↓

1. Create a structural map of the industry/market (I/M) landscape in which the company is embedded
2. Perform a five (or six) forces analysis of the I/M landscape
3. Determine the attractiveness (as measured by the profit potential, the Return on Invested Capital [ROIC]) of the industry
4. Determine or decide on the company's position or competitive strategy.
5. What are the relationships between the players in the I/M landscape.

Step 3 of the SPSP : Execute the plan

Step 1 : Create a structural map

(e.g. Personal Computer Industry)

1980s

